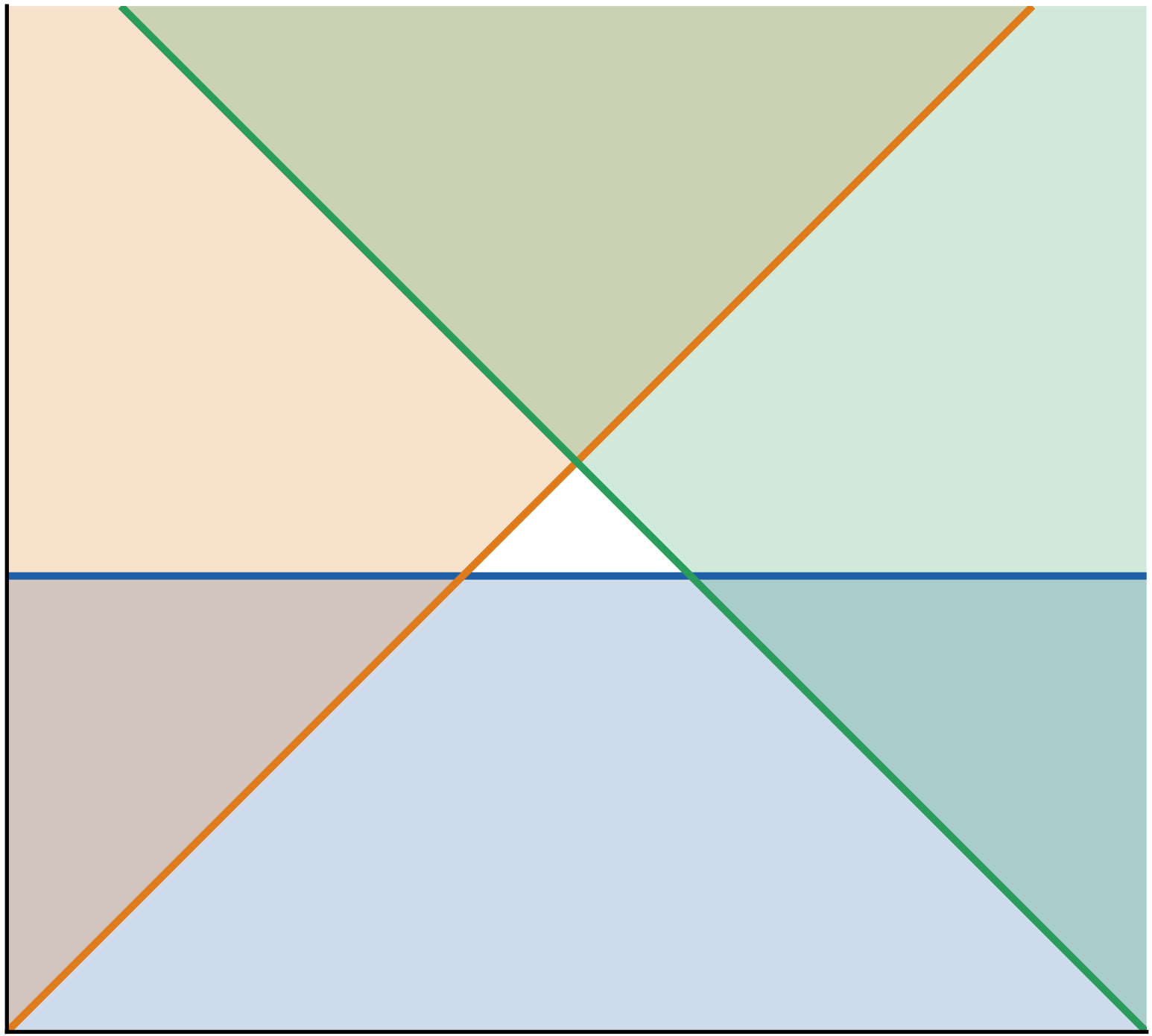Any 2 overlap, all 3 do not:
$n + 1$ cannot be improved